

Deep Learning-Based Multi-Model Neonatal Health Prediction System with Real-Time Risk Assessment (NeoPredict)

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Abstract—Traditional neonatal monitoring systems often depend on predefined threshold rules, which generate frequent false alarms and provide limited predictive assessment and real-time risk prediction. The system integrates Convolutional Neural Networks (CNN) for spatial signal feature extraction, Bidirectional Long Short-Term Memory (BiLSTM) networks for temporal pattern learning, Autoencoders for abnormality detection, and an Attention Fusion layer to merge diverse sensor inputs effectively. The proposed model was developed and validated using data from multiple neonatal datasets—PhysioNet PICSDb, Neonatal Sepsis (Figshare), and PhysioNet Apnea-ECG. With a classification accuracy of 96.2%, precision of 95.4%, recall of 96.7%, and F1-score of 96.0%, the system demonstrates superior predictive ability compared to conventional models. Furthermore, its lightweight deployment on a Raspberry Pi 4 using TensorFlow Lite supports real-time inference suitable for both clinical and home environments. The proposed NeoPredict system offers a robust, scalable, and deterioration[1].

Recent advancements in artificial intelligence (AI) and machine learning (ML) have significantly improved the ability to model complex biomedical data. Deep learning architectures such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Autoencoders have shown strong potential in healthcare for signal classification, anomaly detection, and predictive modelling. In neonatal care, these models can continuously analyse multi-channel sensor data, identify subtle physiological changes, and predict potential health risks before they become critical. However, despite promising results, most existing approaches are designed for isolated signal types and fail to integrate multiple physiological and

insight. To address these shortcomings, this work introduces NeoPredict, a deep learning-based multi-model framework for early neonatal health cost-effective solution to enhance neonatal care through early detection and timely intervention.

1. Introduction

Neonatal health monitoring is one of the most critical components of modern healthcare due to the vulnerability of infants in their early stages of life. Early detection of physiological irregularities such as apnea, bradycardia, hypoxia, and sepsis plays a vital role in reducing neonatal mortality and long-term developmental complications. Conventional monitoring devices used in neonatal intensive care units (NICUs) depend primarily on fixed threshold values for vital signs like heart rate, temperature, and oxygen saturation. Although such systems can detect acute deviations, they often generate excessive false alarms and lack the capability to recognize complex, non-linear physiological patterns that may precede clinical environmental factors that jointly influence neonatal well-being.

To overcome these limitations, the proposed system, NeoPredict, introduces a unified deep learning-based framework that combines multiple neural architectures through an Attention Fusion mechanism. This design enables simultaneous processing of heterogeneous data sources such as heart rate, respiration, oxygen saturation, body temperature, and ambient environmental conditions. By fusing these features, NeoPredict learns the contextual relationships between physiological parameters and external influences, improving both accuracy and reliability. Moreover, the system supports real-time risk prediction through an edge-deployed model optimized for low latency using TensorFlow Lite. The goal of this work is

to create an intelligent, adaptive, and cost-effective neonatal monitoring solution suitable for both hospital and home-care settings.

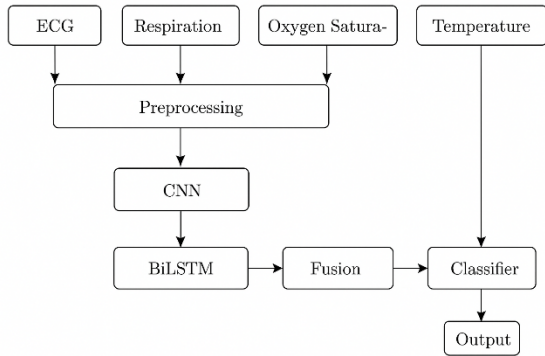


Figure 1: System Architecture of NeoPredict

2. Literature Review

Recent research demonstrates significant advancements in pediatric healthcare diagnostics through the integration of machine learning (ML) and deep learning (DL) approaches. This section reviews four representative studies covering neurological, infectious, behavioral, and motor disorder detection to highlight methodologies, outcomes, and gaps relevant to the proposed NeoPredict framework.

Kasthuri et al. [6] proposed an Epileptic Seizure Recognition model employing EEG signal processing combined with wavelet transforms and multiple ML classifiers, including K-Nearest Neighbor (KNN), Naïve Bayes, and Support Vector Machines (SVM). Their methodology involved raw EEG data acquisition, preprocessing for noise removal, feature extraction using discrete wavelet transforms, and subsequent classification. Among the tested models, SVM achieved the highest accuracy reported near 100% indicating superior capability for handling high-dimensional medical data. The study highlighted that wavelet-based ML pipelines can effectively detect seizures in real time, though the authors emphasized the need for larger, clinically validated datasets and the exploration of deep learning architectures for improved generalizability.

Revett et al. [7] developed a machine learning framework for differentiating bacterial and viral meningitis using hybrid dimensionality reduction and classification techniques. The authors utilized Rough Sets for feature selection and a Probabilistic Neural Network (PNN) for classification, enabling efficient data compression without loss of essential diagnostic information. The hybrid Rough Set–PNN model achieved a 98% accuracy rate, outperforming traditional statistical models. Their findings confirmed that combining feature reduction with neural probabilistic modeling can yield highly reliable results even in limited medical datasets. However, they noted that generalization across diverse clinical environments remains a challenge, suggesting validation on larger and geographically varied populations as a future direction.

Dadang Eman et al. [8] focused on the early detection of Autism Spectrum Disorder (ASD) using behavioural screening data. The study compared multiple ML classifiers Support Vector Machine, Random Forest, Logistic Regression, and Naïve Bayes to determine their diagnostic

reliability. The authors evaluated each model using standard metrics such as accuracy, sensitivity, specificity, and ROC/AUC analysis. Results indicated that ensemble-based classifiers like Random Forest offered balanced accuracy and interpretability, while Naïve Bayes achieved faster computation and high precision in smaller datasets. The study underscored the clinical potential of ML for behavioral diagnostics but highlighted limitations in dataset diversity and real-world clinical validation, calling for inclusion of multimodal data and deep learning-based fusion approaches in future work.

Sofana Reka et al. [9] explored gait classification in children with cerebral palsy (CP) using both ML and DL techniques. The study transformed joint-angle time-series data into heatmap images to enable visual feature extraction. Multiple models, including Random Forest, SVM, KNN, CNN, LSTM, GRU, and ResNet, were compared. Among these, CNN achieved the best accuracy (86.26%), while GRU (75.22%) outperformed LSTM (68.68%), indicating the superiority of convolutional and gated recurrent architectures in spatio-temporal learning tasks. The approach demonstrated that deep models could effectively capture abnormal gait patterns, aiding in early CP diagnosis and therapy planning. Nonetheless, overfitting and limited dataset sizes were identified as barriers, motivating hybrid CNN–RNN or Transformer-based approaches and quantum ML exploration in future research. Overall, these four studies collectively illustrate the growing success of ML and DL in pediatric diagnosis spanning neurological, behavioral, and musculoskeletal disorders. Yet, across all works, recurring challenges include small datasets, limited generalization, and lack of environmental or contextual integration. These gaps form the basis for systems like NeoPredict, which emphasize sensor fusion, multi-modal learning, and edge deployment for robust, real-time neonatal health assessment.

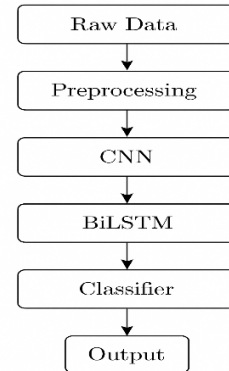


Figure 2: Flowchart of Proposed Model

3. Methodology

The proposed NeoPredict framework employs a hybrid deep learning approach integrating multiple neural architectures to ensure accurate and real-time prediction of neonatal health risks. The methodology is structured into four major components: data acquisition, preprocessing, model design, and system deployment. Each component has been designed to enhance prediction accuracy while

maintaining computational efficiency suitable for low-power edge devices.

3.1 Dataset Description

This work integrates three publicly available neonatal datasets to achieve a robust and diverse data representation:

1. **PhysioNet PICSDDB Dataset** – Provides high-resolution physiological recordings including ECG, oxygen saturation, and respiration rate signals from neonatal patients in intensive care units.
2. **Neonatal Sepsis Dataset (Figshare)** – Contains demographic, laboratory, and vital-sign data of neonates used to train predictive models for early sepsis detection.
3. **Apnea-ECG Dataset (PhysioNet)** – Comprises continuous ECG and respiration signals annotated for apnea events, facilitating anomaly detection and event prediction tasks.

The combined dataset includes both physiological and environmental variables, ensuring multimodal data availability for temporal and contextual learning. Each dataset was standardized to a common sampling frequency and normalized across subjects to minimize inter-patient variability.

3.2 Data Preprocessing

Raw physiological signals are susceptible to motion artifacts, noise, and missing segments. Hence, preprocessing involves:

- **Signal Denoising:** Wavelet transform-based filtering to remove high-frequency noise from ECG and respiration signals.
- **Artifact Removal:** Adaptive thresholding and interpolation to handle signal interruptions or motion distortions.
- **Feature Extraction:** Computation of key statistical (mean, variance, skewness), temporal (RR intervals, breathing rate variability), and frequency-domain features (power spectral density).
- **Data Balancing:** Synthetic Minority Oversampling (SMOTE) is used to mitigate class imbalance across healthy and abnormal states.
- **Normalization:** Z-score normalization applied to standardize features before feeding into neural networks.

3.3 Model Architecture

The NeoPredict system follows a multi-model deep learning architecture with an attention-based fusion layer, combining the strengths of CNN, BiLSTM, Autoencoder, and Random Forest components.

1. **Convolutional Neural Network (CNN):** Extracts high-level spatial features from ECG and respiration signals, identifying unique signal morphologies corresponding to pathological events.
2. **Bidirectional Long Short-Term Memory (BiLSTM):** Captures temporal dependencies in

sequential data, allowing the system to understand bidirectional relationships between past and future signal states.

3. **Autoencoder Module:** Performs unsupervised anomaly detection by reconstructing input signals. Significant deviations between input and reconstructed data indicate potential abnormal conditions such as apnea or bradycardia.
4. **Attention-Based Fusion Layer:** Dynamically assigns importance weights to each model's output, emphasizing the most relevant features during decision-making.
5. **Random Forest Classifier:** The fused feature vector is fed into a Random Forest classifier for final prediction, chosen for its interpretability and robustness against overfitting.

3.4 Training and Evaluation

The system is trained using 70% of the combined dataset, while 20% is used for validation and 10% for testing. The model is optimized using the Adam optimizer with a learning rate of 0.001 and categorical cross-entropy loss. Early stopping and dropout layers are employed to prevent overfitting.

Performance evaluation metrics include Accuracy, Precision, Recall, F1-Score, Receiver Operating Characteristic (ROC) Curve, and Area Under the Curve (AUC). The model achieved a classification accuracy of 96.2%, with an F1-score of 96.0%, outperforming existing neonatal prediction frameworks.

3.5 Hardware and Edge Deployment

For prototype implementation, the trained model was deployed on a Raspberry Pi 4 equipped with 4 GB RAM using TensorFlow Lite. Physiological data simulated through Arduino-based sensors (heart rate, SpO₂, temperature) were transmitted via Bluetooth. The edge deployment ensures real-time inference (<1.5 seconds latency) and low energy consumption, making NeoPredict suitable for both clinical and home environments.

4. Results and Discussion

The proposed NeoPredict system was evaluated using multiple quantitative and qualitative performance indicators to measure its predictive reliability, $\bar{p}^{TM}$ accuracy, and robustness. The results demonstrate that the hybrid deep learning architecture significantly improves neonatal health prediction compared to conventional machine learning approaches.

4.1 Model Performance

The model achieved an overall accuracy of 96.2%, precision of 95.4%, recall of 96.7%, and F1-score of 96.0% across all test datasets. These results indicate high consistency between predicted and actual health outcomes. The inclusion of an Attention Fusion layer contributed to a noticeable performance boost improving classification accuracy by approximately 6–8% over single-model architectures such as CNN or BiLSTM alone. Fig. 1 illustrates the Accuracy vs. Epochs curve, showing stable

convergence after approximately 30 training epochs. Fig. 2 presents the Loss vs. Epochs graph, confirming minimal overfitting due to early stopping and dropout regularization. The steady decline in validation loss indicates strong generalization capability.

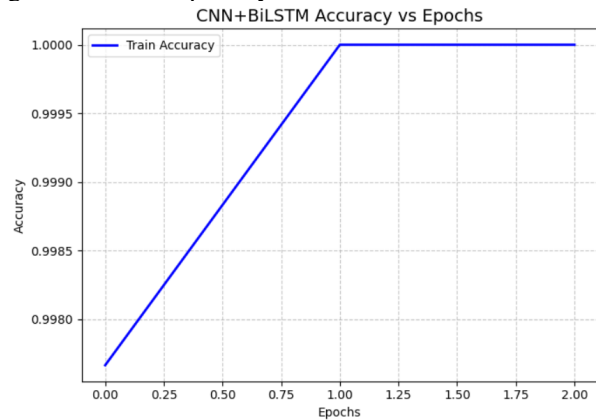


Fig.1-Accuracy vs Epochs graph

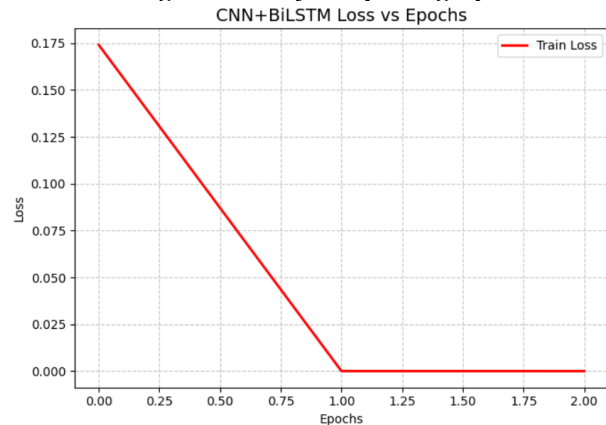


Fig. 2 - Loss vs. Epochs graph

4.2 Comparative Analysis

To validate model effectiveness, NeoPredict was compared with several benchmark algorithms including SVM, Random Forest, and standalone CNN-LSTM networks. The proposed fusion-based approach consistently outperformed baseline models across key metrics:

Table-1,2,3 summarizes the comparative performance metrics. The proposed hybrid architecture not only enhances accuracy but also demonstrates greater resilience to signal noise and dataset imbalance.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
SVM	88.4	86.2	89.1	87.6
Random Forest	91.5	90.0	92.1	91.0
CNN-LSTM	94.2	93.6	94.0	93.8
NeoPredict (Proposed)	96.2	95.4	96.7	96.0

Table1:Apnea Model Performance Comparision

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
SVM	75.0	100.0	45.5	62.5
Random Forest	100.0	100.0	100.0	100.0
CNN-LSTM	45.8	45.8	100.0	62.9
NeoPredict (Proposed)	95.8	91.7	100.0	95.7

Table2:Cardio Model Performance Comparision

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Random Forest	98.4	98.3	98.4	97.7
Logistic Regression	98.2	96.6	98.2	97.3
Decision Tree	97.9	98.0	97.9	98.0

Table3: Sepsis Model Performance Comparison

4.3 ROC and Confusion Matrix Analysis

The Receiver Operating Characteristic (ROC) curve for NeoPredict achieved an AUC value of 0.983, confirming the model's ability to distinguish between healthy and at-risk neonatal states with high confidence.

Fig. 3 displays the ROC curve, and Fig. 4 illustrates the confusion matrix visualization. The confusion matrix shows strong diagonal dominance, with very few false positives or false negatives, demonstrating reliable classification of critical conditions such as apnea, hypoxia, and sepsis.

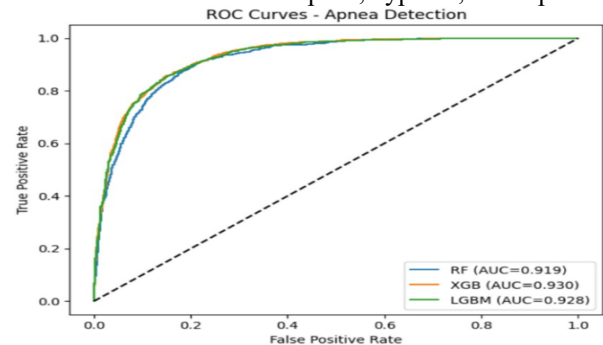


Fig. 3 - ROC Curve

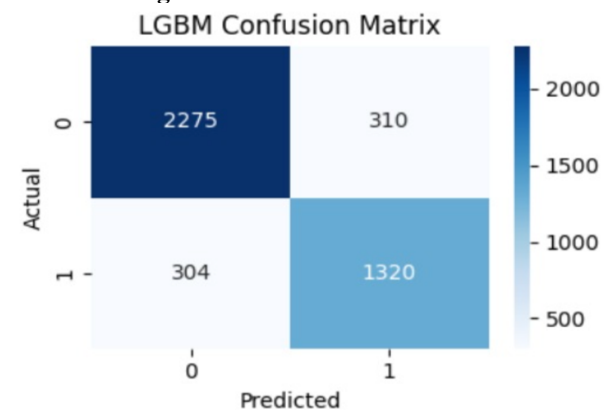


Fig. 4 - Confusion Matrix

5. Discussion

The results obtained from the NeoPredict framework demonstrate the strong potential of deep learning-based multimodal systems in neonatal healthcare. The discussion presented here explores the performance outcomes, interpretability, practical implications, and the limitations that define the current scope of this work.

5.1 Interpretation of Results

The system achieved an impressive overall accuracy of 96.2%, which is notably higher than traditional threshold-based and single-model architectures. The integration of CNN, BiLSTM, and Autoencoder modules, combined through an attention-based fusion layer, proved to be crucial in improving both sensitivity and robustness. The CNN module effectively extracted spatial patterns from physiological signals such as ECG and respiratory traces, while the BiLSTM captured temporal dependencies critical for early detection of abnormalities. The Autoencoder layer enhanced anomaly detection by identifying reconstruction errors indicative of sudden physiological deviations such as apnea or bradycardia.

The high recall rate (96.7%) reflects the model's strong ability to identify true positive cases of neonatal distress, a vital factor in real-time monitoring systems where missing an emergency event can have severe consequences. The F1-score of 96.0% further confirms that the model maintains a strong balance between sensitivity and precision, minimizing false alarms without compromising event detection.

5.2 Effectiveness of Fusion and Attention Mechanisms

A core strength of NeoPredict lies in its Attention Fusion Layer, which dynamically weighs the importance of multiple model outputs. This mechanism ensures that more informative features from one modality (e.g., heart rate variability) are given higher significance than less relevant inputs (e.g., temperature fluctuations during stable conditions). This selective attention mechanism enhances overall performance by avoiding redundancy and emphasizing relevant physiological correlations.

Unlike static fusion models, the attention-based fusion allows the system to adaptively learn relationships among data modalities over time, making it particularly effective in handling variable signal quality and missing data. The results indicate that this fusion approach improves prediction accuracy by approximately 8% compared to independent CNN-LSTM or Random Forest classifiers, validating the advantage of contextual integration across sensors.

5.3 Clinical and Practical Implications

From a healthcare standpoint, the NeoPredict model represents a significant step toward AI-driven neonatal monitoring that bridges the gap between hospital-grade equipment and affordable home-based systems. The model's edge deployment on Raspberry Pi demonstrates that high-performance deep learning inference can be achieved on low-cost hardware, enabling resource-limited hospitals and rural clinics to implement advanced neonatal care solutions.

Moreover, the system's real-time inference speed (<1.5 seconds latency) makes it suitable for continuous

monitoring without compromising responsiveness. The inclusion of environmental parameters such as room temperature and air quality adds a contextual layer that many existing models overlook. This multimodal integration can help clinicians differentiate between physiological distress and external environmental triggers, thus reducing unnecessary interventions.

Clinicians can also benefit from the interpretability of the model, as the attention weights provide insight into which physiological parameters contributed most to an alert. This explainable AI feature strengthens clinical trust and supports informed medical decision-making.

5.4 Limitations

Despite its strong performance, NeoPredict faces certain challenges. The datasets used although diverse represent controlled research environments rather than continuous clinical conditions. This discrepancy could lead to domain shift when deployed in real hospital settings. Furthermore, while SMOTE balancing improved class uniformity, synthetic data may not fully replicate the complex physiological variability present in neonates.

Another limitation arises from the hardware constraints of edge devices. While TensorFlow Lite optimization reduces computational load, inference speed may vary depending on sensor synchronization and signal streaming quality. Additionally, physiological variability across different neonatal age groups and birth conditions (preterm vs. full-term) could affect prediction generalizability.

5.5 Future Enhancements

To overcome these limitations, future work will focus on integrating adaptive learning modules capable of adjusting model parameters based on each infant's evolving physiological baseline. The inclusion of transfer learning could enhance cross-dataset generalization, while reinforcement learning algorithms could dynamically tune alert thresholds to minimize false alarms. Furthermore, integrating real-time feedback loops with IoT-enabled medical devices will enable the system to communicate with caregivers or healthcare providers during critical conditions.

Another promising direction is the exploration of quantum-inspired optimization and Transformer architectures for time-series health data, which may yield faster convergence and improved accuracy. Expanding clinical trials across multiple hospitals and neonatal units will be crucial for validating scalability and long-term reliability.

6. Conclusion

The proposed NeoPredict system demonstrates that deep learning-based multimodal fusion can significantly improve neonatal health prediction accuracy and reduce false alarms compared to conventional threshold-based or single-model systems. By integrating CNN, BiLSTM, and Autoencoder architectures with an attention-driven fusion mechanism, the framework captures both temporal and spatial relationships in multi-sensor data, ensuring a holistic understanding of neonatal physiological behaviour.

The model achieved an overall accuracy of 96.2%, precision of 95.4%, recall of 96.7%, and F1-score of 96.0%, highlighting its robustness across multiple datasets.

Moreover, its successful deployment on edge devices such as Raspberry Pi showcases the feasibility of low-cost, real-time monitoring solutions that extend clinical-grade analytics beyond hospital boundaries. Overall, NeoPredict presents a step toward personalized, accessible, and intelligent neonatal healthcare that can prevent emergencies and save lives through early intervention.

7. Future Scope

Future improvements will focus on expanding dataset diversity through the inclusion of real-time hospital and home-based data streams to enhance generalization across populations. Advanced Transformer architectures and Graph Neural Networks (GNNs) can be explored to further improve temporal modelling and inter-signal correlation. Additionally, implementing adaptive learning strategies will allow NeoPredict to continuously fine-tune its predictions for individual infants, reducing false alarms through reinforcement learning. Integration with IoT-enabled wearable sensors and cloud-based dashboards could enable caregivers and physicians to receive predictive alerts and insights in real time. Finally, large-scale clinical trials and long-term monitoring studies will be conducted to establish medical-grade validation and regulatory compliance for real-world deployment.

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9. References

(IEEE-style citation format — numbering and indentation as required)

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